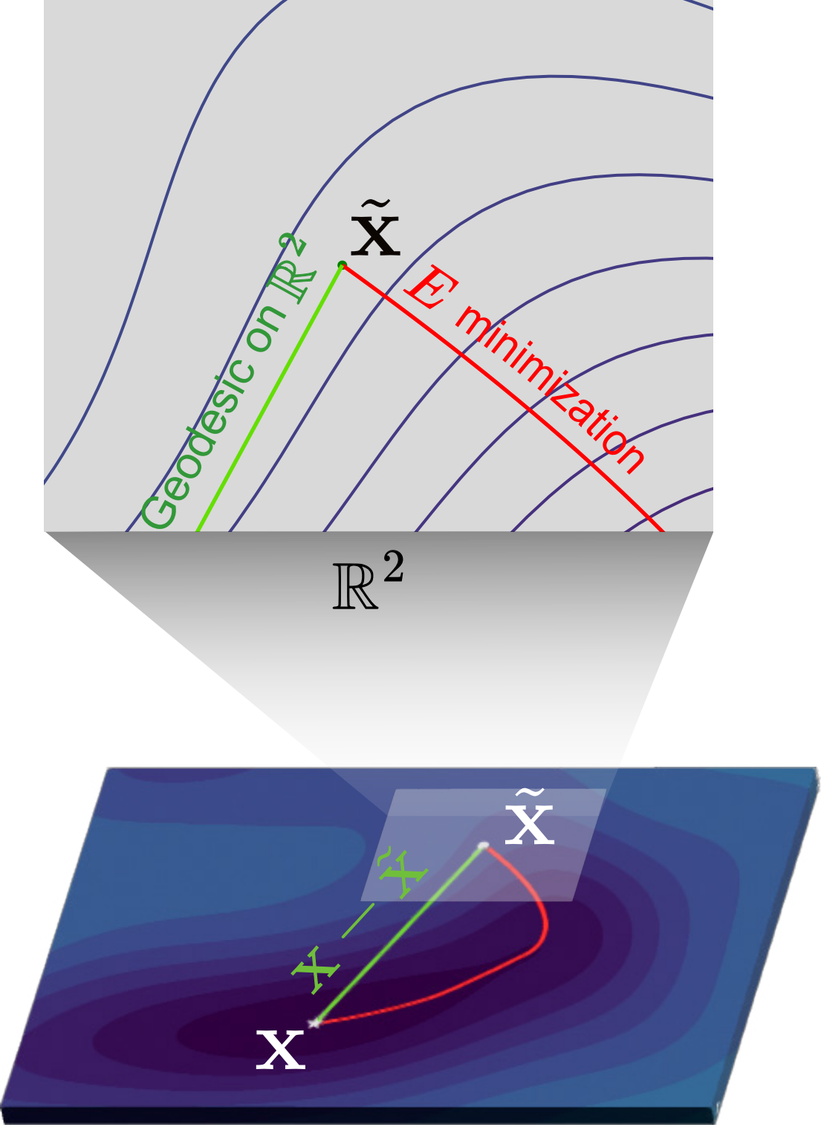
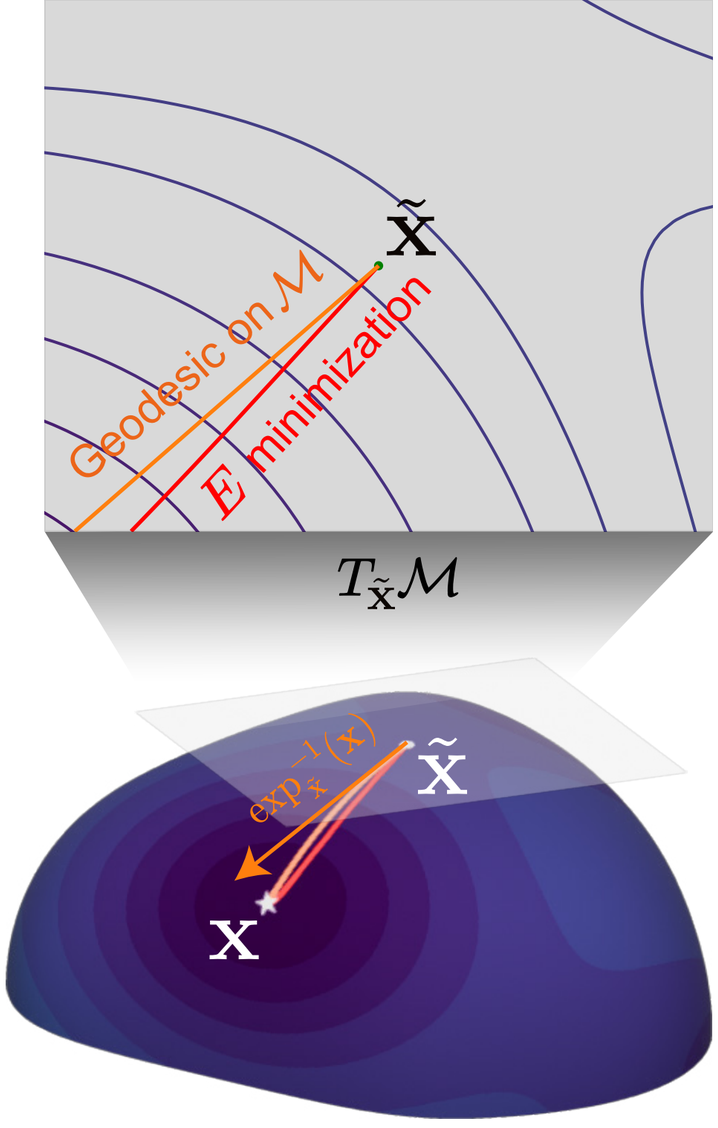
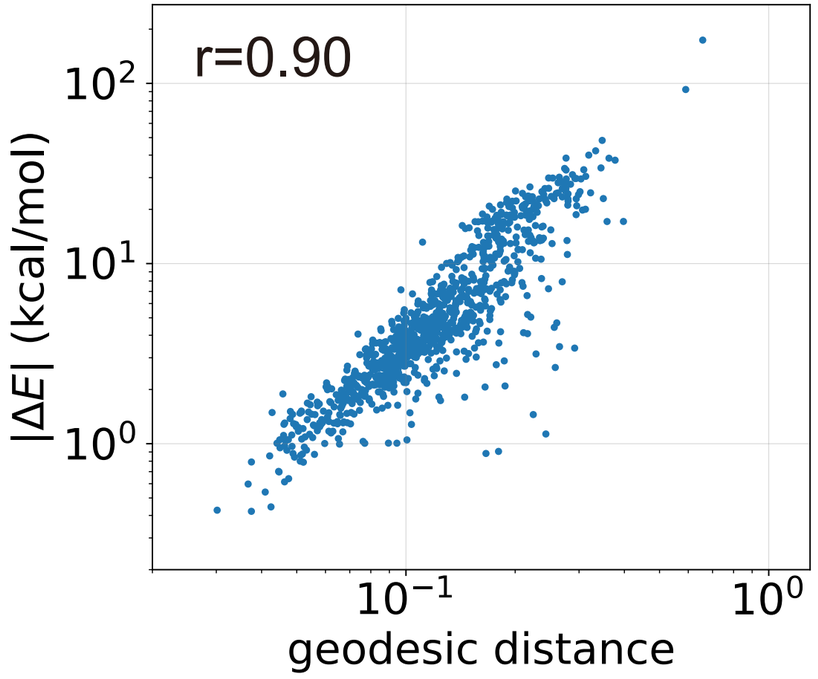
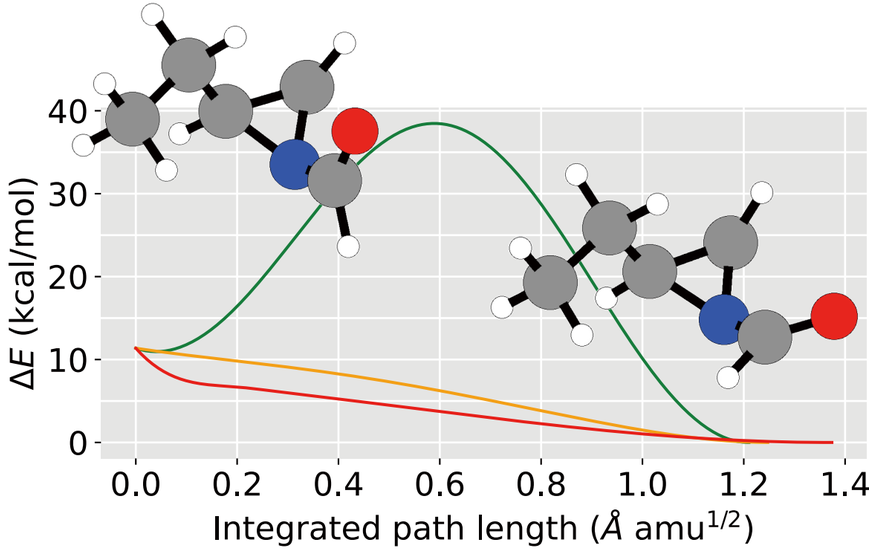


Euclidean view $\mathbb{R}^2$



Riemannian manifold view $\mathcal{M}$